

Challenges in Indian Agriculture and Government Interventions: A Review

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Abstract- India's agriculture sector remains the backbone of rural livelihoods and national food security, contributing substantially to economic growth and employment. However, farmers continue to encounter a wide range of structural and socio-economic barriers, including small and fragmented landholdings, heavy reliance on monsoon rains, inadequate technological adoption, post-harvest inefficiencies, financial vulnerabilities, and unstable market prices. Additional constraints such as rising labour expenses, low levels of mechanization, limited irrigation coverage, and insufficient knowledge of sustainable practices further limit agricultural productivity. This review paper explores these complex challenges in detail while assessing the effectiveness of major government programmes designed to address them. Key schemes—such as the Pradhan Mantri Fasal Bima Yojana (PMFBY), PM-Kisan income support, Soil Health Card initiative, e-NAM digital marketplace, Pradhan Mantri Krishi Sinchai Yojana (PMKSY), Minimum Support Price (MSP) mechanisms, and emerging digital agriculture efforts—are evaluated for their role in improving productivity, farmer income, and risk management. The study identifies notable policy successes as well as areas requiring improvement, emphasizing the need for integrated, technology-oriented, and farmer-focused strategies.

Keywords – Indian agriculture, Farmer income, Government interventions, Climate change, Digital divide, Agricultural policy, Rural livelihoods, Sustainability.

I. INTRODUCTION

Agriculture remains a cornerstone of India's socio-economic system, despite its declining contribution to national output. While the sector accounted for nearly 50–55% of Gross Domestic Product (GDP) in the 1950s, its share fell to approximately 15–17% in 2023–24 due to structural economic transformation. However, agriculture continues to employ about 45–46% of the workforce, implying that nearly 200 million people depend on a sector generating a relatively small share of national income. This persistent employment–output imbalance reflects low labour productivity and underemployment in rural areas.

Structural constraints significantly contribute to this imbalance. Nearly 85–86% of operational holdings are small or marginal (below two hectares), limiting mechanization, investment, and economies of scale. Rising input costs—growing at 6–10% annually—combined with price volatility and post-harvest losses of 15–20%, further weaken farm profitability. These challenges are exacerbated by environmental stress, as nearly 60% of cultivated land is rainfed, making agricultural output highly vulnerable to climate variability, groundwater depletion, and soil degradation.

Although digital technologies and precision agriculture are promoted as solutions, their adoption remains uneven. Less than 40% of rural households have reliable internet access, and limited digital literacy constrains the use of digital platforms, particularly among smallholders. Labour dynamics also pose challenges; large-scale rural–urban migration and rising real wages have led to seasonal labour shortages. With over 90% of agricultural workers engaged informally, productivity improvements remain limited, and cultivation costs for labour-intensive crops have increased by 20–30%.

In response, the Government of India has implemented multiple interventions focusing on income support, risk mitigation, irrigation expansion, and market reforms. While these initiatives have improved coverage and reduced certain risks, their impacts remain uneven due to gaps in implementation, awareness, and institutional coordination.

Against this backdrop, this paper reviews the key challenges confronting Indian agriculture and critically evaluates major government interventions. The study synthesizes evidence from existing research to identify structural bottlenecks, assess policy effectiveness, and outline pathways toward inclusive, resilient, and sustainable agricultural development.

Fig. 1 illustrates an integrated framework of the major drivers contributing to low income and profitability in Indian agriculture. Climate and environmental risks, labour and workforce constraints, market and supply chain inefficiencies, digital divide, and institutional gaps interact cumulatively to increase production costs, income volatility, and livelihood insecurity. The framework highlights that these challenges are interdependent rather than isolated, underscoring the need for coordinated, multi-sectoral policy interventions to improve farm incomes and agricultural sustainability.

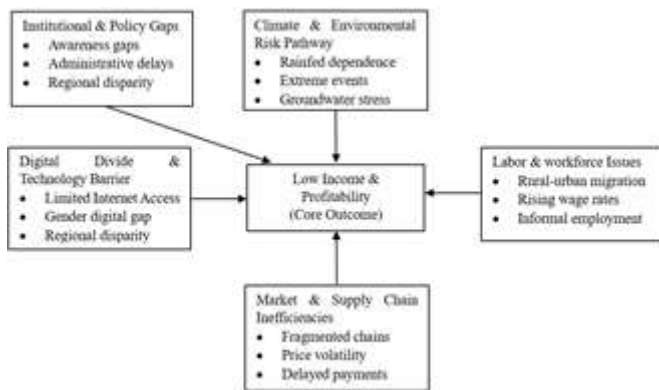


Fig. 1. Integrated framework of key challenges affecting Indian agriculture.

Fig. 2 illustrates the macro-level structure of challenges confronting Indian agriculture, highlighting the imbalance between high workforce dependence (about 45–46%) and low GDP contribution (15–17%), alongside widespread smallholder dominance and rainfed cultivation. The surrounding components depict interconnected constraints related to income and profitability, digital access, climate risk, labour dynamics, market inefficiencies, and institutional gaps. Together, the figure emphasizes the systemic and interdependent nature of agricultural challenges, underscoring the need for integrated policy responses rather than isolated interventions.



Fig. 2. Core Agricultural Indicators: Employment Share, GDP Contribution, and Structural Constraints

II. LITERATURE REVIEW

Guided by the integrated framework presented in Fig. 1, the literature review systematically examines each challenge dimension and synthesizes empirical and policy-oriented evidence related to Indian agriculture.

A. Low Income and Profitability

Despite employing nearly half of India's workforce, agriculture contributes a relatively small share to national GDP, reflecting persistent underemployment, low labour productivity, and deep-rooted structural inefficiencies within the sector [1], [2], [22]. Evidence from national surveys indicates that the average monthly income of agricultural households remains substantially lower than that of non-agricultural households, reinforcing income inequality and livelihood insecurity in rural areas [23]. This widening income disparity has been associated with increasing farmer indebtedness, distress-driven migration, and growing reliance on government support mechanisms [6], [14].

Rising cultivation costs—including fertilizers, certified seeds, diesel, irrigation, and hired labour—have consistently outpaced growth in output prices, leading to sustained pressure on net farm incomes [6]. Empirical analyses of farm-size economics reveal that small and marginal farmers, who account for over 85% of total landholdings, face structural disadvantages arising from land fragmentation, limited access to institutional credit, and weak bargaining power in output markets [22], [29]. In addition, deficiencies in storage, logistics, and cold-chain infrastructure contribute to substantial post-harvest losses, estimated at nearly ₹92,000 crore annually, further undermining farm profitability and overall food system efficiency [7], [24]. Uneven implementation of Minimum Support Price (MSP) procurement across crops and regions further constrains income protection, particularly for farmers producing non-MSP or highly perishable commodities [25].

TABLE I. Key Driving Factors Contributing to Low Income and Profitability in Indian Agriculture

Sr. No.	Segment of Driving Factor	Driving Factor	Details of Driving Factor
1	Socio-Economic	Low labour productivity	High workforce participation with low output contribution results in low per-capita farm income indebtedness among small and marginal farmers [1], [2], [22].
2	Economic	Rising input costs	Increasing costs of fertilizers, seeds, fuel,

			irrigation, and labour reduce profit margins [6], [24], [29]
3	Socio-Economic	Fragmented landholdings	Small farm sizes restrict mechanization, economies of scale, and bargaining power [22], [29]

TABLE I summarizes the key socio-economic and economic drivers contributing to low income and profitability in Indian agriculture. Socio-economic constraints, such as low labour productivity and fragmented landholdings, limit output per worker, restrict economies of scale, and weaken farmers' bargaining power, especially among small and marginal landholders. Simultaneously, economic pressures arising from escalating input costs erode profit margins and heighten financial vulnerability. Collectively, these interrelated factors explain the persistent income instability and indebtedness observed among farming households and highlight the structural nature of profitability challenges in Indian agriculture. Addressing these issues requires coordinated policy interventions focused on productivity enhancement, cost rationalization, land consolidation mechanisms, and strengthened market support systems.

B. Digital Divide and Technology Barriers

Digital agriculture initiatives such as e-NAM, AgriStack, and Kisan Suvidha have been introduced to improve market transparency, price discovery, and access to advisory services. However, the effectiveness of these initiatives remains constrained by uneven rural internet penetration, limited smartphone ownership, and low levels of digital literacy across states and socio-economic groups, particularly among smallholders and women farmers [8], [9]. Empirical studies indicate that inadequate awareness, language barriers, limited trust in digital platforms, and insufficient training significantly reduce farmer engagement with digital marketplaces and advisory systems [10].

Disparities in digital infrastructure across states further widen the technology adoption gap, with farmers in rainfed, tribal, and geographically remote regions deriving the least benefit from digital reforms [21]. Evidence from international experiences suggests that in the absence of inclusive infrastructure, farmer-centric platform design, and sustained capacity-building efforts, technology-driven agricultural interventions may reinforce existing socio-economic inequalities rather than alleviate them [30]. As a result, the persistent digital divide continues to limit the adoption of precision agriculture, data-driven decision-making, and climate-smart farming practices in India [27].

TABLE II. Major Drivers Underlying the Digital Divide and Technology Barriers in Indian Agriculture

Sr. No.	Segment of Driving Factor	Driving Factor	Details of Driving Factor
1	Technological	Limited rural internet access	Poor connectivity restricts use of digital platforms and advisory services [8], [9], [21]
2	Socio-Economic	Gender digital gap	Women farmers have reduced access to smartphones and digital tools [9], [10], [30]
3	Infrastructure	Regional infrastructure disparity	Remote and rainfed regions benefit least from digital reforms [21], [27]

TABLE II presents the major drivers underlying the digital divide and technology barriers in Indian agriculture. Technological constraints, particularly limited rural internet access, restrict farmers' ability to effectively use digital platforms, online marketplaces, and real-time advisory services. Socio-economic factors, notably the gender digital gap, further intensify inequalities, as women farmers often face reduced access to smartphones, digital tools, and skill development opportunities. In addition, infrastructure-related disparities across regions mean that farmers in remote and rainfed areas benefit least from digital innovation initiatives. Collectively, these factors constrain the adoption of digital agriculture technologies and weaken the overall effectiveness of technology-enabled government interventions, underscoring the need for inclusive digital infrastructure expansion, targeted capacity-building programs, and gender-sensitive policy design.

C. Climate and Environmental Challenges

A substantial proportion of Indian agriculture remains dependent on rainfall, with nearly 60% of the cultivated area classified as rainfed. This high dependence on monsoon patterns makes agricultural output particularly sensitive to climatic variability and extreme weather events, including droughts, floods, heatwaves, and unseasonal rainfall [11]. Climate projections indicate that, in the absence of effective adaptation and mitigation measures, yields of major staple crops could decline by 10%–25% by the mid-21st century [12], [20]. The growing frequency and intensity of extreme weather events have already resulted in recurrent crop failures, income volatility, and an increased reliance on risk-mitigation instruments such as crop insurance under schemes like PMFBY [16].

Beyond climatic variability, environmental degradation has intensified production risks and constrained long-term sustainability. Excessive extraction of groundwater for irrigation has caused significant declines in water tables across several agricultural regions, threatening irrigation reliability and increasing energy costs for farmers [13]. Soil-related challenges—including nutrient imbalance, declining organic matter, and land degradation—have reduced soil fertility and heightened dependence on chemical inputs, thereby raising production costs and environmental externalities [23]. In addition, climate-induced shifts in pest and disease dynamics have increased crop losses and input requirements, further undermining farm resilience, particularly among small and marginal farmers with limited adaptive capacity [11], [12].

TABLE III. Environmental and Climatic Drivers Affecting Agricultural Sustainability in India

Sr. No.	Segment of Driving Factor	Driving Factor	Details of Driving Factor
1	Climatic Vulnerability	Rainfed agriculture dependence	Heavy reliance on monsoon rainfall increases vulnerability to climate variability [11], [12]
2	Climate Risk	Extreme weather events	Droughts, floods, and heatwaves cause crop failures and income shocks [12], [16], [20]
3	Natural Resource Stress	Groundwater depletion	Over-extraction threatens irrigation sustainability and raises costs [13], [23]

TABLE III summarizes the major climatic and environmental drivers affecting agricultural sustainability in India. Climatic vulnerability arising from rainfed agriculture exposes farmers to rainfall uncertainty and yield instability. Climate risks associated with extreme weather events contribute directly to crop damage and income shocks. Simultaneously, natural resource stress, particularly groundwater depletion, threatens the long-term viability of irrigated agriculture by increasing costs and reducing water security. Together, these interconnected factors underscore the urgency of strengthening climate-resilient farming practices, sustainable resource management, and adaptive policy frameworks to safeguard agricultural productivity and livelihoods.

D. Labour and Workforce Issues

Structural transformation of the Indian economy and sustained rural–urban migration have significantly altered the agricultural labour landscape. While agriculture still employs nearly 45–46% of India’s total workforce, its share in total employment has declined steadily over the past two decades,

leading to acute labour shortages during peak sowing and harvesting seasons [14]. Census and labour survey data indicate that more than 30% of rural male workers aged 18–35 have migrated to non-farm employment, reducing the availability of able-bodied agricultural labour at the village level. As a result, real agricultural wage rates have increased by 6–8% annually in recent years, raising cultivation costs and disproportionately affecting labour-intensive crops such as paddy, sugarcane, cotton, and horticulture [15].

Despite gradual increases in mechanization, Indian agriculture remains highly labour-dependent. Informal employment dominates the sector, with over 90% of agricultural workers engaged without written contracts, job security, or social protection, limiting productivity gains and workforce stability [26]. Gender disparities further intensify workforce challenges. Women account for nearly 40–45% of agricultural labour input, yet earn 20–30% lower wages than male counterparts and hold less than 15% of operational land titles, restricting their access to institutional credit, mechanization services, and extension support [15], [26]. Policy initiatives promoting mechanization, custom hiring centres, and collective farming through Farmer Producer Organizations (FPOs) aim to offset labour shortages and improve efficiency. However, adoption remains uneven, as high capital costs, limited technical skills, and institutional constraints continue to exclude a large share of small and marginal farmers, who constitute over 85% of total landholdings [21], [27].

TABLE IV. Labour and Workforce-Related Drivers Impacting Agricultural Productivity

Sr. No.	Segment of Driving Factor	Driving Factor	Details of Driving Factor
1	Labour & Demographic	Rural–urban migration	Labour shortages during peak seasons increase cultivation costs [14], [15]
2	Labour & Economic	Rising wage rates	Higher wages adversely affect labour-intensive crops [15], [26]
3	Institutional	Informal labour arrangements	Lack of formal contracts limits productivity and social security [26], [27]

Table IV summarizes the key labour and workforce-related drivers affecting agricultural productivity. Labour and demographic factors, particularly rural–urban migration, have reduced workforce availability during critical agricultural operations, leading to seasonal labour shortages and increased production costs. Economic drivers, reflected in rising wage rates, further intensify financial pressure on farmers cultivating

labour-intensive crops. Institutional drivers, such as the predominance of informal labour arrangements, limit productivity growth and restrict access to social security, skill development, and mechanization support. Collectively, these factors underscore the need for integrated policy interventions focused on appropriate mechanization, labour formalization, gender-inclusive workforce strategies, and targeted skill enhancement to ensure productivity growth and long-term sustainability in Indian agriculture.

D. Market Access and Supply Chain Inefficiencies

Despite policy reforms aimed at improving agricultural marketing, farmers continue to face limited direct access to markets due to fragmented supply chains and dependence on intermediaries. Inadequate transportation networks, warehousing facilities, and cold storage infrastructure reduce farmers' bargaining power and result in distress sales, particularly for perishable commodities such as fruits, vegetables, and dairy products [7], [16]. Price volatility, delayed payments, and lack of transparent pricing mechanisms further discourage market participation and private investment in productivity-enhancing technologies [25].

Although digital platforms such as e-NAM seek to integrate agricultural markets and improve price discovery, their impact remains constrained by uneven state-level implementation, limited trader participation, interoperability issues, and inadequate last-mile logistics support [8], [16]. As a result, the expected gains from market integration and competition have not been fully realized, limiting improvements in farmer income and market efficiency [21].

TABLE V. Market and Supply Chain Drivers Limiting Farmer Income Realization

Sr. No.	Segment of Driving Factor	Driving Factor	Details of Driving Factor
1	Supply Chain Structure	Fragmented supply chains	Multiple intermediaries reduce farmer price realization [7], [16], [21]
2	Price Mechanism	Price volatility	Unstable prices discourage investment and planning [25]
3	Transaction & Payment Systems	Delayed payments	Cash-flow constraints weaken farmer participation in markets [16], [25]

The factors summarized in Table V capture structural, pricing, and transactional drivers that constrain effective market participation by farmers. Weak supply chain structures,

characterized by multiple intermediaries, reduce farmers' share in consumer prices and limit income realization. Inefficiencies in price mechanisms, manifested through high volatility, introduce uncertainty that discourages production planning and private investment. In addition, shortcomings in transaction and payment systems, particularly delayed payments, impose liquidity constraints that weaken sustained market engagement. Collectively, these drivers hinder market efficiency and farmer profitability, highlighting the need for integrated supply-chain reforms, price risk management instruments, and timely payment mechanisms.

E. Institutional and Policy Implementation Gaps

India has introduced numerous government schemes targeting income support, risk mitigation, irrigation expansion, and market reforms. However, gaps in implementation, awareness, and inter-agency coordination continue to limit their effectiveness. Studies highlight delays in benefit disbursement, exclusion and inclusion errors, and regional disparities in scheme coverage, disproportionately affecting small and marginal farmers [2], [17], [28].

Weak institutional capacity at the local level, limited reach of agricultural extension services, and inadequate monitoring and evaluation mechanisms further constrain policy outcomes [4], [19]. As a result, the translation of policy intent into measurable improvements in farm productivity, income stability, and climate resilience remains inconsistent across states and agro-climatic regions [27], [30].

TABLE VI. Institutional and Policy-Related Drivers Affecting Agricultural Development Outcomes

Sr. No.	Segment of Driving Factor	Driving Factor	Details of Driving Factor
1	Information & Outreach	Scheme awareness gaps	Limited awareness reduces access to government benefits [2], [17], [28]
2	Administrative Capacity	Administrative delays	Slow disbursement weakens scheme impact [4], [19]
3	Governance & Federal Coordination	Regional disparities	Uneven implementation across states affects outcomes [27], [30]

The institutional drivers presented in Table VI influence the translation of policy intent into on-ground outcomes. Limitations in information and outreach restrict farmers' awareness of available schemes, reducing effective coverage. Constraints in administrative capacity, reflected in procedural delays and slow fund disbursement, weaken the timeliness and impact of interventions. Furthermore, governance and federal coordination challenges contribute to regional disparities in

implementation, resulting in uneven development outcomes across states. Together, these institutional shortcomings underscore the need for improved outreach mechanisms, streamlined administrative processes, and more consistent inter-state coordination to enhance policy effectiveness.

III. CONCLUSION

This paper reviewed the key structural, economic, technological, environmental, and institutional challenges affecting Indian agriculture and examined the role of major government interventions in addressing these constraints. The analysis indicates that fragmented landholdings, rising input costs, labour shortages, climate variability, limited market access, and digital adoption barriers continue to suppress productivity and income growth, particularly among small and marginal farmers. These challenges remain interlinked and require coordinated responses rather than isolated policy measures.

The review further shows that government initiatives related to income support, crop insurance, irrigation development, market reforms, and digital platforms have contributed to risk mitigation and improved access to services in selected regions. However, their overall effectiveness is limited by implementation gaps, uneven regional coverage, and institutional constraints at the local level. Strengthening last-mile delivery, enhancing farmer awareness, improving extension systems, and integrating technology with on-ground infrastructure are essential to improve policy outcomes. A more integrated and evidence-driven approach is required to achieve sustainable productivity growth and income stability in Indian agriculture.

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